Safe Multi-Agent Graph Learning for Fast Frequency Response of Virtual Power Plants in Inverter-Based Islanded Microgrids

Tran Kim Nhat, Nguyen Van Cuong, *Graduate Student Member, IEEE*, Nguyen Phi Long, *Member, IEEE*, and Laurent El Ghaoui

***Abstract*—High penetration of inverter-based resources (IBRs) in islanded microgrids lowers inertia and makes fast frequency response (FFR) sensitive to feeder reconfiguration, which the fixed droop control, topology blind droop loop cannot track. When IBRs are aggregated into virtual power plants (VPPs) that bid FFR reserve into the ancillary service market, the committed reserve is only as deliverable as the active feeder can route, so the VPP's real-time execution layer must stay topology aware as the distribution system operator reconfigures the grid. We propose a topology adaptive FFR controller for this layer: An inductive GraphSAGE encoder over the active feeder graph feeds a parameter shared multi-agent proximal policy optimization actor under centralized training with decentralized execution, and each resource jointly emits a fast power reference and a local droop gain inside a shared safety envelope. Trained on one topology of a modified IEEE 123-bus feeder and evaluated zero-shot on 24 unseen reconfigurations, the proposed method achieves the lowest unseen integral of time-weighted absolute error among all standards-compliant methods. The learning baselines and fixed droop control yield errors 1.45–1.75 times and 3.7 times higher, respectively, while the proposed method exhibits no degradation beyond the training feeder and delivers the lowest-cost FFR service, reducing the average distribution system operator procurement cost to €1.87 per event, 10–13% below that of the best competing learning controllers.**

***Index Terms*—Fast Frequency Response, Islanded Microgrids, Inverter-based Resource, Virtual Power Plant, GraphSAGE Network, Multi-Agent Reinforcement Learning**

I. Introduction

THE transition toward 100% inverter-based resources (IBRs) in islanded distribution feeders removes the synchronous inertia that has traditionally bounded the rate of change of frequency (RoCoF), pushing the frequency nadir after a disturbance toward underfrequency load shedding (UFLS) thresholds and, in severe cases, blackout [1]. Secure operation therefore requires the distributed energy resources (DERs) interfaced through inverters, namely battery energy storage system (BESS), vehicle-to-grid (V2G) chargers, and distributed photovoltaics (DPV), to deliver fast frequency response (FFR) on subsecond to second time scales, coordinated with the grid-forming (GFM) inverters that set the voltage and frequency reference [2]. In market operation these DERs do not act in isolation: they are aggregated into virtual power plants (VPPs) that bid energy and fast frequency reserve into the distribution ancillary service market coordinated by a distribution system operator (DSO) [3], so the FFR loop studied here is the real-time execution layer through which a VPP delivers the reserve it has committed. The scheme deployed today is the classical droop controller, in which each DER injects an active power increment proportional to the local frequency deviation through a fixed per device gain. This loop has two structural limitations. First, the traditional controller is fixed: commissioned for the worst foreseeable event, it is underutilized during mild disturbances and cannot amplify near UFLS or reallocate headroom when storage energy is scarce [4]. Second, the controller is topology blind: FFR deliverability is set jointly by the DER fleet and the active feeder graph, which DSO tie switch reconfiguration and line contingencies continually reshape [5], yet neither the gain nor the controller state encodes the active graph.

Adaptive virtual inertia and virtual synchronous generator (VSG) control improves RoCoF and nadir over fixed droop and can stabilize a 100% IBR microgrid without rotating machines [6], but is retuned by small signal analysis around a fixed feeder, treating topology variation as parametric uncertainty rather than a structural change of the graph. Model-based optimization and model predictive control can handle operational constraints well but heavily depend on model accuracy. Additionally, constantly re-solving the optimization problem as conditions change creates a high computational burden, limiting real-time implementation [7]. Multi-agent reinforcement learning (MARL) under centralized training with decentralized execution (CTDE) coordinates large DER populations for dispatch, voltage, and primary frequency control [8], [9], yet standard formulations embed a fixed adjacency through multilayer perceptron (MLP) or transductive graph convolution encoders [10], [11] and expose either a power reference or a droop gain, rarely both [12]. Closest to this setting, a learning-based VPP frequency regulation controller with transient safety guarantees tolerates a reconfiguration by recomputing participation factors and retraining for large changes [13], and techno-economic frameworks integrate fast response reserve into VPP ancillary service operation [14]; both aggregate IBRs on a given network, so a topology change is met by parameter reallocation or retraining. The present controller instead encodes the active graph with an inductive GraphSAGE encoder [15], [16] and is evaluated zero-shot across 24 reconfigurations with no retraining. To our knowledge no prior study combines an inductive encoder, a

The authors acknowledge the financial support provided by the Vingroup Innovation Foundation (VINIF) under Project VINIF.2025.DA119. *(Corresponding author: Nguyen Phi Long.)*

T. K. Nhat, N. V. Cuong, N. P. Long, and L. El Ghaoui are with Center for Environmental Intelligence and College of Engineering & Computer Science, VinUniversity, Hanoi 100000, Vietnam (e-mails: nhat.tk@vinuni.edu.vn; 25cuong.nv@vinuni.edu.vn; long.np2@vinuni.edu.vn; laurent.eg@vinuni.edu.vn).

N. V. Cuong and N. P. Long are also with School of Electrical and Electronic Engineering, Hanoi University of Industry, Hanoi 100000, Vietnam (e-mails: cuongnv@haui.edu.vn; long.nguyen04@haui.edu.vn).

per DER joint power reference and droop gain action, and an islanded 100% IBR setting under a zero-shot topology holdout; the novelty lies in the problem, architecture, and evaluation, not in GraphSAGE itself [15].

Three gaps follow: reserve deliverability is modeled from local energy alone and counts network-inaccessible reserve as deliverable; power references and droop gains are designed separately rather than jointly; and controllers are evaluated only on their training topology, leaving robustness to reconfiguration unquantified. This paper develops a topology adaptive FFR controller for islanded 100% IBR microgrids that learns over the active feeder graph, exposes a joint power reference and droop gain action, and is verified for zero-shot transfer across feeder reconfigurations. Three contributions follow.

1) **Topology coupled deliverability and frequency model.** Reserve deliverability is modeled as a function of the active feeder graph, bus voltages, and inverter loading, while closed-loop frequency dynamics are represented by a topology-dependent reduced-order model. Reconfiguration therefore affects both reserve accessibility and the synchronizing and disturbance-coupling structure of the frequency response, eliminating the inconsistency of combining topology-aware control with topology invariant for dynamic grid.
2) **GraphSAGE-MAPPO with a joint action and a safety envelope.** An inductive GraphSAGE encoder over the active feeder graph provides node embeddings to a parameter-shared multi-agent proximal policy optimization (MAPPO) actor under CTDE. The actor outputs a power reference and droop gain for each DER. Frequency-security constraints are addressed by a safety envelope: a monotonic-gain parameterization with a Common Lyapunov stability certificate, a backbone-governed RoCoF bound, and a closed-form one-step projection that holds the predicted next-step center of inertia (COI) within the band.
3) **Zero-shot topology holdout on frequency security metrics.** The policy is trained on a single feeder topology and tested without retraining on 24 unseen reconfigurations of a modified islanded IEEE 123-bus feeder. An encoder-only ablation, replacing GraphSAGE with an MLP while keeping the remaining learning architecture unchanged, isolates the contribution of the graph encoder to topology adaptation. To the best of our knowledge, this is the first MARL-based FFR controller for islanded microgrids validated through zero-shot topology transfer.

II. System and Frequency Response Models

A. Islanded 100% Inverter Based Microgrid and Component Models

1) Grid-Forming Inverters: A backbone of n_g GFM inverters establishes the voltage and frequency references. Each unit g emulates the swing dynamics of a synchronous machine through a virtual inertia control loop [17],

$$2H_g \frac{d\Delta\omega_g}{dt} = \Delta P_{\mathrm{set},g} - \Delta P_{e,g} - D_g\,\Delta\omega_g, \tag{1}$$

$$\frac{d\Delta\delta_g}{dt} = \omega_0\,\Delta\omega_g, \tag{2}$$

where $\Delta P_{e,g}$ is the network imposed electrical power deviation introduced below, $\omega_0 = 2\pi f_0$ is the nominal angular frequency, and the virtual inertia H_g is a programmable control parameter rather than rotating mass [18]. The backbone is heterogeneous (one VSG plus n_g-1 droop GFM units; per unit values in Table I). The grid-following (GFL) fleet acts only over the finite step Δt, contributing to the per step RoCoF and to the post-nadir damping mechanism described in Sec. III-D.

2) Grid-Following DER Fleet: A fleet of $|\mathcal{A}|$ GFL DERs, including BESS, V2G chargers, and DPV, injects active power with a learned droop term but no inertia. These controllable agents constitute the VPP's FFR layer, which is the only layer designed here. The GFM backbone provides inertial response and primary droop, while a proportional–integral AGC loop restores the steady-state offset over minutes. At the one-second fast step Δt, the inner voltage and current loops behave as first-order envelopes. DPV applies curtailment-only regulation, and each BESS/V2G unit operates as a state-of-charge (SoC) integrator with charge/discharge efficiencies $\eta_{\mathrm{ch}}, \eta_{\mathrm{dis}}$, energy capacity E, and interval ΔT [19],

$$\mathrm{SoC}_{i,t+1} = \begin{cases} \mathrm{SoC}_{i,t} - \dfrac{P_{i,t}\,\Delta T}{\eta_{\mathrm{dis}}\,E_i}, & P_{i,t} > 0 \text{ (discharge)}, \\ \mathrm{SoC}_{i,t} + \dfrac{|P_{i,t}|\,\eta_{\mathrm{ch}}\,\Delta T}{E_i}, & P_{i,t} < 0 \text{ (charge)}, \end{cases} \tag{3}$$

held within physical and FFR-eligibility SoC bands; the V2G fleet is time-varying through connection availability and departures.

The reserve a DER can actually deliver is not a function of this energy state alone. In an islanded feeder it also depends on the active topology and the resulting voltage and current envelopes,

$$R^{\mathrm{ffr}}_{i,t} = f(\mathcal{G}_t,\ \mathrm{SoC}_{i,t},\ V_t,\ S_t,\ P_t,\ Q_t)\,, \tag{4}$$

where V_t, S_t, P_t, Q_t are the bus voltage, apparent loading, and active and reactive line-flow profiles of the energized feeder. A controller that ignores $\mathcal{G}_t$ commits reserve that cannot be delivered under the realized topology, producing undersupply and larger frequency excursions; this topology-dependent deliverability is the physical basis of the first contribution.

B. Market and Execution Context

The DER fleet is organized for market participation by partitioning the feeder into $|\mathcal{Z}|$ VPP zones with agent sets $\mathcal{A}_z$, where $\mathcal{A} = \bigcup_{z\in\mathcal{Z}} \mathcal{A}_z$. As illustrated in Fig. 1, this work designs and evaluates only the fast execution layer, while treating the upper market and network-management layers as exogenous operating context. The DSO provides the active feeder topology and associated network signals through its operational scheduling process [20], while the VPP aggregator supplies the committed energy and reserve targets obtained from its market-clearing procedure under uncertainty [21], [22]. The proposed controller acts as a price-taking execution layer that converts these signals into real-time active-power and droop commands, consistent with the hierarchical structures commonly adopted in distribution-level flexibility markets [23].

The fast-frequency reserve commitment is represented by the deliverable quantity $R^{\text{ffr}}_{i,t}$ in (4). Consequently, the active feeder graph assumed during reserve commitment must also be observed by the real-time execution layer. Otherwise, topology-blind control may commit reserve that cannot be physically routed to the disturbance location, leading to reserve under-delivery. This coupling motivates the topology-dependent frequency response model developed next.

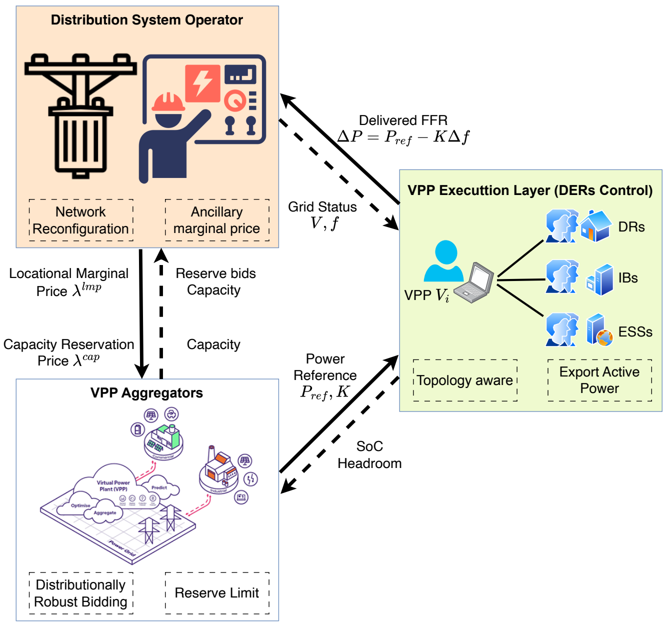

Fig. 1. Operating context: the proposed controller as the DER execution layer (price taker) under DSO and VPP signals.

C. Topology Dependent Frequency Model

The classical COI model collapses the feeder to a single scalar swing equation and therefore cannot express how tie switch reconfiguration or line contingencies redistribute the frequency response. We instead retain the inter unit angle structure and let the active feeder graph enter the state matrix explicitly, recovering the COI quantity only as a readout.

1) Topology-Dependent Couplings via Kron Reduction: The active feeder graph $\mathcal{G}_t$ enters the model through two blocks of one Kron reduction. Partitioning the buses into the retained GFM set I and the eliminated load set L and condensing the latter [24], [25] gives the Kron-reduced active-power Jacobian,

$$\tilde{J}_r = J_{II} - J_{IL}\, J_{LL}^{-1}\, J_{LI}, \tag{5}$$

which sets the synchronizing torque coupling among the GFM units; tie switch reconfiguration and line contingencies change the block partition in (5), so each topology produces a distinct $\tilde{J}_r$. The disturbance enters through the other block of the same partition: the imbalance ΔP_{imb} is injected at the physical feeder bus of the contingency, not spread uniformly across units, so condensing that injection onto the retained units through the Kron transfer $J_L = J_{IL} J_{LL}^{-1}$ makes the input matrix topology dependent,

$$B_{\text{net}}(\mathcal{G}_t) = \operatorname{diag}\!\left(\tfrac{1}{2H}\right) J_L(\mathcal{G}_t), \tag{6}$$

reducing to a uniform, location-agnostic injection only in the degenerate case $J_L \propto \mathbf{1}$. A contingency electrically close to unit g loads it more than distant units, and a reconfiguration that reroutes around g redistributes that loading, so the active graph shapes both the synchronizing coupling and where the disturbance is felt.

2) Reduced-Order State-Space Model: Collect the GFM deviations into the state $x = \left[\Delta\boldsymbol{\delta}_{\text{rel}} \in \mathbb{R}^{n_g-1},\ \Delta\boldsymbol{\omega} \in \mathbb{R}^{n_g}\right]^{\top}$, the relative angles being referenced to the largest rated GFM unit. Assembling (1) and (2) over the backbone gives the linear, topology-parameterized dynamics,

$$\frac{dx}{dt} = A_f(\mathcal{G}_t, K_t)\, x + B_{\text{ref}}\, \Delta P_{\text{ref,t}} + B_{\text{net}}(\mathcal{G}_t)\, \Delta P_{\text{imb}}, \tag{7}$$

where $\Delta \mathbf{P}_{\text{ref},t} \in \mathbb{R}^{n_g}$ is the per-GFM reference vector obtained by aggregating the DER-level scalar commands $\Delta P_{m,t}$ to their nearest GFM units, $\Delta P_{\text{imb},t}$ is the scalar contingency magnitude routed by $B_{\text{net}}(\mathcal{G}_t)$ of (6), and the block state matrix is

$$A_f(\mathcal{G}_t, K_t) = \begin{bmatrix} \mathbf{0} & \omega_0\, C \\ -\operatorname{diag}\!\left(\tfrac{1}{2H}\right) \tilde{J}_r^{\text{rel}}(\mathcal{G}_t) & -\operatorname{diag}\!\left(\tfrac{1}{2H}\right) D(K_t) \end{bmatrix}, \tag{8}$$

with $C \in \mathbb{R}^{(n_g-1)\times n_g}$ the reference differencing operator (row i holds $+1$ at unit i and -1 at the reference unit), $\tilde{J}_r^{\text{rel}}(\mathcal{G}_t) \in \mathbb{R}^{n_g \times (n_g-1)}$ the Jacobian $\tilde{J}_r$ of (5) with its reference column removed, and $D(K_t) = \operatorname{diag}(K_{\text{eff},g})$ the effective damping set by the learned droop gains K_t. The learned per-DER gains, absolute quantities in MW/Hz, enter the per-unit damping after aggregation to the nearest GFM unit g,

$$K_{\text{eff},g} = K_{\text{bb}} + \frac{f_0}{S_{\text{base}}} \sum_{m:\, g(m)=g} K_m, \tag{9}$$

where K_{bb} is the backbone droop and K_m is the learned gain of (15); parallel droop sources add, so the learned GFL droop adds to the backbone damping in $D(K_t) = \operatorname{diag}(K_{\text{eff},g})$.

3) Discretization, Readout, and Secondary Control: Because $\Delta t = 1\,\text{s}$ is large relative to the inverter time constants, the model is discretized by Zero-Order Hold (ZOH) to (Φ, Γ), cached per topology and droop gain bin. The COI frequency is recovered as a rating weighted readout,

$$\Delta f_t = f_0 \frac{\sum_g S_g\, \Delta\omega_g}{\sum_g S_g}, \qquad \dot{f}_t = \frac{\Delta f_t - \Delta f_{t-1}}{\Delta t}, \tag{10}$$

so the COI is an observable of the reduced dynamics rather than the state itself, and is also the input to a slow integral secondary loop that removes the steady state offset. Unlike the single COI swing model, the closed loop carries no aggregate damping or droop constant; damping and droop enter per unit through $D(K_t)$, formed by the backbone droop and the learned GFL droop. The only lumped aggregate is the rating weighted backbone inertia H_{sys}, set by control because H_g is programmable virtual inertia; its numerical value is reported in Sec. IV.

4) COI Metric and Topology Sensitivity: Frequency security is evaluated on the COI readout (10), the system wide frequency observable at the one second step and consistent with reduced order system frequency response (SFR) models [26]. Projecting the swing block onto the rating weights decouples a single real COI mode,

$$\lambda_{\mathrm{COI}} \approx -\frac{\sum_g K_{\mathrm{eff},g}}{\sum_g 2H_g}, \tag{11}$$

With a time constant of $\tau \approx 1.5$ s at the backbone gains of Table I, determined by aggregate droop and inertia and thus nearly topology-invariant, the COI trajectory exhibits quasi–first-order behavior. The remaining $2(n_g - 1)$ eigenvalues correspond to inter-machine swing pairs in the 38–345 Hz range for the base feeder and they cancel in the rating-weighted readout, thereby eliminating the oscillatory governor tail characteristic of synchronous machine SFR models. Reconfiguration influences the COI not through this mode but via the located forcing $B_{\mathrm{net}}(\mathcal{G}_t)$ and the deliverable reserve (4); consequently, a model that neglects $\mathcal{G}_t$ yields inaccurate estimates of both.

III. Proposed GraphSAGE-MAPPO Controller

A. FFR as a Cooperative Dec-POMDP

At each fast step the execution layer observes the active feeder graph $\mathcal{G}_t$, the zonal prices, and the realized frequency, and chooses the per-device fast power reference and droop gain $u_{m,t} = (\Delta P_{m,t}, K_{m,t})$ for every $m \in \mathcal{A}$ to hold the COI frequency inside the security band at minimal control effort,

$$\begin{aligned}
\min_{\{u_{m,t}\}} \quad & \mathbb{E}\left[\sum_t \big(|\Delta f_t| + \beta(|\Delta f_t| - \delta)^+\big)\right] \\
\text{s.t.} \quad & x_{t+1} = \Phi_t x_t + \Gamma_{\mathrm{ref},t}\Delta P_{\mathrm{ref},t} + \Gamma_{\mathrm{net},t}\Delta P_{\mathrm{imb},t}, \\
& A_f(\mathcal{G}_t, K_t) \text{ stable}, \\
& |\dot{f}_t| \le \dot{f}_{\mathrm{lim}}, \qquad |\Delta f_t| \le \delta, \\
& 0 \le K_{m,t} \le K_{\max,m}, \\
& |\Delta P_{m,t}| \le R^{\mathrm{ffr}}_{m,t}(\mathcal{G}_t), \qquad \mathrm{SoC}_{m,t} \in [\underline{s}, \bar{s}],
\end{aligned} \tag{12}$$

where δ is the safe band, $\dot{f}_{\mathrm{lim}}$ is the RoCoF ride-through limit, and $(x)^+ = \max\{0, x\}$. The compact notation $\Phi_t = \Phi(\mathcal{G}_t, K_t)$, $\Gamma_{\mathrm{ref},t} = M(\mathcal{G}_t, K_t)B_{\mathrm{ref}}$, and $\Gamma_{\mathrm{net},t} = M(\mathcal{G}_t, K_t)B_{\mathrm{net}}(\mathcal{G}_t)$ The ZOH-discretized state and input maps define the ZOH discretization of (7).. Because $\mathcal{G}_t$, the disturbance, and the prices are revealed only in real time and the per-device limits are time varying and nonconvex, we do not solve (12) online; instead we cast it as a cooperative decentralized partially observable Markov decision process (Dec-POMDP) $\langle \mathcal{A}, \mathcal{S}, \{O_m\}, \{U_m\}, P, r, \gamma \rangle$ and learn a safety-filtered policy under CTDE. The 41 GFL agents observe only local and neighborhood information; the fast step is $\Delta t = 1\,\mathrm{s}$ with 300 steps per episode, and the architecture is shown in Fig. 2. The objective of (12) maps onto the reward (17) and its constraints onto the safety envelope of Sec. III-D.

1) Observation: Each agent m observes a 16 dimensional local vector comprising the local frequency deviation and RoCoF from the COI readout, net power, SoC and dischargeable energy headroom, the zonal locational marginal price, a four way device type one hot encoding, VPP membership, the previous droop gain and power reference, SoC violation flags, a persistence power forecast, and a departure proxy. The local vectors are placed on the 123 bus graph at their host buses and augmented with four per bus grid background features (active and reactive load, voltage magnitude, and agent density), giving the full feeder observation $obs_{\mathrm{full}} \in \mathbb{R}^{123\times 20}$.

2) Action: The scheme it replaces is the classical droop controller, in which each DER injects a power increment proportional to the local frequency deviation through a fixed per device gain K_i,

$$\Delta P^{\mathrm{ffr}}_{i,t} = -K_i\,\Delta f_t. \tag{13}$$

Each agent instead emits a two dimensional action $a_m = [a^P_m, a^K_m] \in [-1, 1]^2$ that jointly sets a fast active power reference and a local droop gain, replacing this fixed droop gain control with a learned, state conditioned one,

$$\Delta P_m = \eta_P\, a^P_m\, P_{\mathrm{rated},m}, \tag{14}$$

$$K_m = K_{\min,m} + \tfrac{1}{2}\big(1 + a^K_m\big)\big(K_{\max,m} - K_{\min,m}\big), \tag{15}$$

where η_P is the power channel fraction of the device rating. The two channels combine into the realized per DER injection

$$\Delta P^{\mathrm{ffr}}_{m,t} \;=\; \Delta P_m \;-\; K_m\,\Delta f_t, \tag{16}$$

a feedforward reference plus a learned proportional (droop) feedback on the COI readout; the classical loop (13) is recovered as the special case $\Delta P_m = 0$ with K_m fixed. The power channel spans up to $\eta_P = \pm 10\,\%$ of rated power (clipped to the device rating and SoC headroom, rate limited by $\Delta a_{\max} = 0.2$ per step); the droop channel spans $[K_{\min,m}, K_{\max,m}]$ with $K_{\min,m} = 0$ and per type ceilings listed in Table I. A droop gain is enabled only when SoC_m lies in the FFR eligibility band, nested inside the physical band; otherwise $K_m = 0$. Aggregation of the per DER commands to the host GFM units, including the conversion of the droop channel into $D(K_t)$, follows (9). Exposing a^P_m and a^K_m jointly gives two degrees of freedom per device, a feedforward reference to position storage ahead of an FFR call and a state conditioned droop gain to shape the response, neither available to a power only or droop only law.

3) Reward: A single team reward couples frequency security, control effort, and service quality,

$$r_t = r^{\Delta f}_t + r^{\mathrm{nadir}}_t + r^{\mathrm{viol}}_t + r^{\mathrm{ffr}}_t + r^{\mathrm{eff}}_t + r^{\mathrm{soc}}_t + r^{\mathrm{mkt}}_t + r^{\mathrm{commit}}_t. \tag{17}$$

Let $(x)^+ = \max\{0, x\}$ be the positive part, $\Pi_{[a,b]}(y) = \min\{b, \max\{a, y\}\}$ the projection onto $[a, b]$, and $\langle g_m \rangle = \frac{1}{|\mathcal{A}|}\sum_{m\in\mathcal{A}} g_m$ the fleet average. Each security term shares one normalized, deadbanded form

$$e(x; d, x_{\mathrm{ref}}) = \Pi_{[0,1]}\Big(\frac{(|x| - d)^+}{x_{\mathrm{ref}}}\Big), \tag{18}$$

in which the deadband d suppresses penalties from noise near f_0 and the event scaled reference x_{ref} keeps the gradient from saturating across the contingency classes. With $f = f_0 + \Delta f_t$ and alarm threshold f_{th}, the three groups are

$$r_t^{\Delta f} = -\, w_{\Delta f}\, e\big(\Delta f_t;\, d_f, \Delta f_{\text{ref}}\big), \tag{19}$$

$$r_t^{\text{nadir}} = -\, w_{\text{nad}}\, \Pi_{[0,1]}\Big(\tfrac{(f_{\text{th}}-f)^+ + \big(f-(2f_0-f_{\text{th}})\big)^+}{\Delta f_{\text{nad}}}\Big), \tag{20}$$

$$r_t^{\text{viol}} = -\, w_{\text{viol}}\, \Pi_{[0,2]}\Big(\tfrac{(|\Delta f_t|-\delta)^+}{\delta}\Big), \tag{21}$$

$$r_t^{\text{ffr}} = +\, w_{\text{ffr}}\, \Pi_{[0,1]}\big(1 - |\Delta f_t|/\delta\big), \tag{22}$$

$$r_t^{\text{eff}} = -\, \frac{w_{\text{eff}}}{a_{\text{ref}}^2} \left\langle \|a_m\|^2 + \kappa\, \|\Delta a_m\|^2 \right\rangle, \tag{23}$$

$$r_t^{\text{commit}} = -\, w_{\text{com}} \left\langle (\Delta K_m / K_{\max,m})^2 \right\rangle, \tag{24}$$

$$r_t^{\text{soc}} = -\, w_{\text{soc}} \left\langle [(\underline{s} - \text{SoC}_m)^+]^2 + [(\text{SoC}_m - \bar{s})^+]^2 \right\rangle, \tag{25}$$

$$r_t^{\text{mkt}} = +\, w_{\text{mkt}}\, \Pi_{[-1,1]} \big\langle \tilde{\lambda}^{\text{lmp}}\, a_m^P + \tilde{\lambda}^{\text{cap}}\, K_m / K_{\max,m} \big\rangle, \tag{26}$$

where $\tilde{\lambda}^{\text{lmp}} = \lambda^{\text{lmp}}/\lambda^{\text{lmp}}_{\text{ref}}$ and $\tilde{\lambda}^{\text{cap}} = \lambda^{\text{cap}}/\lambda^{\text{cap}}_{\text{ref}}$ are the energy and capacity prices normalized by their references, $(\underline{s}, \bar{s})$ is the usable SoC band, and the per term weights w, deadbands d, references x_{ref}, together with a_{ref} and κ in Sec. IV-B. The first group $(r^{\Delta f}, r^{\text{nadir}}, r^{\text{viol}})$ realizes the security objective of (12) and applies identically to every controller, including the baselines; the second $(r^{\text{ffr}}, r^{\text{eff}}, r^{\text{commit}})$ rewards in band delivery and regularizes effort and reserve smoothness; the third $(r^{\text{soc}}, r^{\text{mkt}})$ ties the policy to the SoC limits and the prevailing reserve price. The reward stream is passed through a running normalizer and clipped to $[-10, 10]$.

The 20 mHz frequency deadband sits between the european network of transmission system operators for electricity (ENTSO-E) maximum frequency containment reserve (FCR) deadband (10 mHz) and the typical AGC restoration deadband (30 mHz), suppressing nuisance gradients from measurement scale fluctuations while preserving sensitivity to genuine events.

B. Inductive GraphSAGE Encoder

The encoder is the component that makes the controller transferable to unseen feeder configurations. We use an inductive GraphSAGE encoder [27], which learns aggregator functions over node neighborhoods rather than a fixed per node embedding. For node j at layer k,

$$H_j^{(k)} = \sigma\Big(W_{\text{self}}\, H_j^{(k-1)} + W_{\text{neigh}} \cdot \text{mean}_{u \in \mathcal{N}(j)}\, H_u^{(k-1)}\Big), \tag{27}$$

where $H_j^{(0)}$ is the per bus observation, σ the rectified linear unit nonlinearity, $\mathcal{N}(j)$ the neighbors of j in the active graph, and $W_{\text{self}}, W_{\text{neigh}}$ the self and neighbor aggregator weights shared across all nodes and topologies; two layers map the 20 dimensional input through a 128 dimensional hidden layer to a 128 dimensional embedding. Because $\mathcal{N}(j)$ is rebuilt at every step from the active graph $\mathcal{G}_t$ and the learned weights belong to the aggregator rather than to a fixed adjacency, reconfiguration changes which neighbors are aggregated but not the aggregating function. This is what enables zero-shot transfer: a transductive graph convolution ties its spectral filters to one adjacency and must be retrained when the graph changes, whereas the inductive encoder satisfies the topology coupled deliverability relation without per topology retraining. After two passes over the 123 bus graph, the embeddings at the 41 DER host buses feed the actor and the centralized critic, both reasoning over a representation that already encodes the realized feeder's neighborhood structure.

C. Multi-agent Proximal Policy Optimization under CTDE

The controller is trained with MAPPO [28] under CTDE: a centralized critic uses global information during training while each agent executes from its local observation alone at run time, and both heads operate on the per agent embeddings from the encoder above show in Fig. 2.

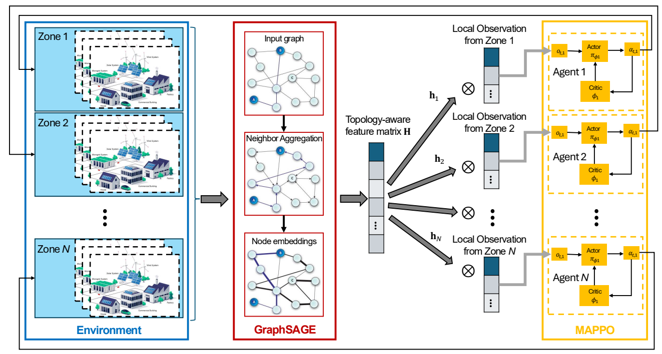

Fig. 2. Proposed GraphSAGE-MAPPO controller.

The actor is a single parameter shared diagonal Gaussian policy that maps each agent embedding through a two layer perceptron to the mean and log standard deviation of the two dimensional action, with bounded mean $\mu = \bar{a}\, \tanh(\cdot)$ and spread $\sigma = \varsigma(\cdot) + \sigma_0$, where ς is the softplus and $\bar{a} = 0.75,\ \sigma_0 = 0.05$. Parameter sharing keeps the policy size independent of $|\mathcal{A}|$ and permits a varying agent count across topologies. The centralized critic concatenates each agent embedding with a global context formed by mean and max pooling over all agents and returns a per agent value, retaining credit for each DER while the pooled context supplies the system level signal for cooperative coordination.

For the update, advantages follow generalized advantage estimation (GAE) ($\gamma = 0.99,\ \lambda = 0.95$) [29] on the shared team reward (17), and the policy is updated by the clipped proximal policy optimization (PPO) surrogate with ratio ρ_t and clip range $\epsilon = 0.2$ [30]; the loss adds a value regression term and an entropy bonus annealed across the curriculum. The complete training procedure is summarized in Algorithm 1.

D. Shared Safety Envelope

Frequency security is enforced by an envelope that belongs to the environment rather than to any controller, so the proposed policy and every baseline act inside the same safe set. The envelope rests on four mechanisms: a nonnegative droop parameterization, a Lyapunov stability certificate, a backbone governed RoCoF bound, and a closed form one-step projection on the predicted COI. The first two guarantee asymptotic stability; the last two bound the transient.

Algorithm 1 Topology-aware MAPPO training for FFR.

Input: Islanded feeder environment with frequency model (7); shared GraphSAGE encoder f_ϕ, Gaussian actor π_θ, centralized critic V_ψ; six phase curriculum $\mathcal{C}$; PPO settings $(\epsilon, \gamma, \lambda, K, B)$

Output: Trained parameters (ϕ, θ, ψ)

```
Initialize (φ, θ, ψ) and the observation/reward normalizers
for each curriculum phase c = 1, ..., 6 do
    Set learning rate, entropy weight, and event distribution from C_c
    for each episode do
        Sample topology G_t and contingency ΔP_imb
        Reset x ← 0
        for t = 1, ..., T do
            z_i ← f_φ(G_t) at each host bus, with the COI skip (Δf_t, ḟ_t)
            Sample (a_i^P, a_i^K) ~ π_θ(· | z_i); form (ΔP_ref, K_t) by (14) and (15)
            Project ΔP_ref for nadir safety; step the dynamics; read out the COI (10)
            Store (z, a, log π, V_ψ, r_t) with r_t from (17)
        end for
        Compute per agent GAE advantages; update (φ, θ, ψ) by the clipped PPO loss
    end for
end for
return (φ, θ, ψ)
```

Stability starts at construction. With $K_{\min} = 0$, the action map (15) keeps the droop gains nonnegative, so $D(K_t) = \mathrm{diag}(K_t) \succeq 0$ and the realized droop is monotone in frequency [13], [31]. The lossless swing-energy argument motivates the certificate in (28), while the lossy case is verified numerically through the LMI in (29). For the swing energy:

$$V = \tfrac{1}{2}\Delta\boldsymbol{\omega}^\top M \Delta\boldsymbol{\omega} + \tfrac{1}{2}\Delta\boldsymbol{\delta}_{\mathrm{rel}}^\top \tilde{J}_r^{\mathrm{sync}} \Delta\boldsymbol{\delta}_{\mathrm{rel}}, \tag{28}$$

with $M = \mathrm{diag}(2H_g)$ and $\tilde{J}_r^{\mathrm{sync}}$ the synchronizing block in relative coordinates, the lossless limit gives $\dot{V} = -\Delta\boldsymbol{\omega}^\top D(K_t)\Delta\boldsymbol{\omega} \le 0$. As retained resistances make $\tilde{J}_r$ nonsymmetric, we instead certify the lossy $A_f(\mathcal{G}_t, K_t)$ numerically by a Common Lyapunov function $P \succ 0$,

$$A_i^\top P + P A_i \preceq -\varepsilon I \quad \text{at every vertex } A_i. \tag{29}$$

Because A_f is affine in K_t, the residual $A_i^\top P + PA_i$ is affine in K_t for any fixed P; a single $P \succ 0$ satisfying (29) at the box vertices therefore certifies $A_i^\top P + PA_i \preceq -\varepsilon I$ over the entire gain box by convex combination. The vertex check is necessary, but the common P not vertex Hurwitzness alone, since the set of Hurwitz matrices is itself non-convex supplies the box-wide guarantee. A common P over the held family would give exponential stability under arbitrary switching [32]; as that linear matrix inequality is infeasible here, stability is instead certified by an average dwell-time bound $\tau_a^\star = \ln\mu/(2\eta) = 41.6$ s, where $\eta = 0.283$ s^{-1} is the slowest strictly-damped decay rate and $\mu = 1.69\times10^{10}$ bounds the inter-mode Lyapunov conditioning over the 96 frozen-time vertices (all Hurwitz). This is met by orders of magnitude since reconfiguration is event-triggered at the minute timescale.

The transient is bounded by the other two mechanisms. The analytic bound $\Delta P_{\mathrm{imb}}/(2H_{\mathrm{sys}})$ is the worst case of a uniform injection scalar model in which the backbone absorbs the imbalance evenly; under the located forcing $B_{\mathrm{net}}(\mathcal{G}_t)$ the disturbance loads the backbone unevenly, so the realized RoCoF can differ from this bound. This first step value is in any case set by the backbone inertia alone and is invariant to the GFL action [33]. A dual-rate framework separates the 1 s control macro-step from the 100 ms simulation micro-step to avoid conflicts. MAPPO and the safety envelope run at the 1 s boundary using cached ZOH matrices. Within each 1 s interval, control loops react instantly to frequency deviations, providing continuous sub-second damping. The 100 ms steps also yield high-resolution data to evaluate peak RoCoF over a 500 ms window for IEEE Std 1547 compliance [34].

The nadir is the limit the controller must shape, one step ahead. With the cached ZOH pair (Φ, M), the ω-block row selection $(\cdot)_\omega$, the COI weights $w_g = S_g/\sum_g S_g$, and the input gain $G = \mathrm{diag}(\frac{1}{2H})$, the next-step COI deviation is affine in the per-GFM reference vector $\Delta\mathbf{P}_{\mathrm{ref},t}$,

$$\Delta f^+ = a + b^\top \Delta P_{\mathrm{ref,t}}, \tag{30}$$

$$a = f_0\, w^\top\big[(\Phi x_t)_\omega + M_{\omega\omega}\, c_u\big], \qquad b = f_0\, G\, M_{\omega\omega}^\top\, w, \tag{31}$$

where $M_{\omega\omega}$ is the ω-block of M and c_u collects the reference-independent inputs (the one-step-delayed AGC feedback and the located disturbance forcing). If $\Delta f^+ \in [-\delta, \delta]$ the action passes unchanged; otherwise it is projected onto the violated half space in closed form,

$$\Delta P_{\mathrm{ref,t}} \;\leftarrow\; \Delta P_{\mathrm{ref,t}} + b\,\frac{\mp\delta - \Delta f^+}{\|b\|^2}, \tag{32}$$

the minimum norm correction satisfying $a + b^\top \Delta P_{\mathrm{ref}} = \mp\delta$ with $\delta = 0.5$ Hz. One linear constraint means no quadratic program is solved in the loop [35], and the guard acts on the COI because that is the observable on which the controller is rewarded and evaluated.

IV. Case Study

A. Test System and Scenarios

The controller is evaluated on a modified IEEE 123 bus distribution feeder [36] operated as a 100% inverter based islanded microgrid (see Fig. 3). Device and system parameters are listed in Table I, and the ENTSO-E frequency security thresholds in Table II.

Four disturbance classes, each scaled as a fraction of S_{base}, are applied (see Table III). Topology generalization is assessed under a strict zero-shot protocol: the policy is trained on a single fixed topology G_0 and evaluated, with no retraining or adaptation, on 24 reconfigurations $\mathcal{G}_{\mathrm{test}} = \{G_1, \ldots, G_{24}\}$ with $G_0 \notin \mathcal{G}_{\mathrm{test}}$, generated from G_0 by tie switch operations and $N-1$ line contingencies. Radial reconfiguration changes only a bounded fraction of edges (Jaccard edge distance up to ≈ 0.18), so the protocol stresses structural generalization within the reachable configuration space rather than transfer to arbitrary graphs.

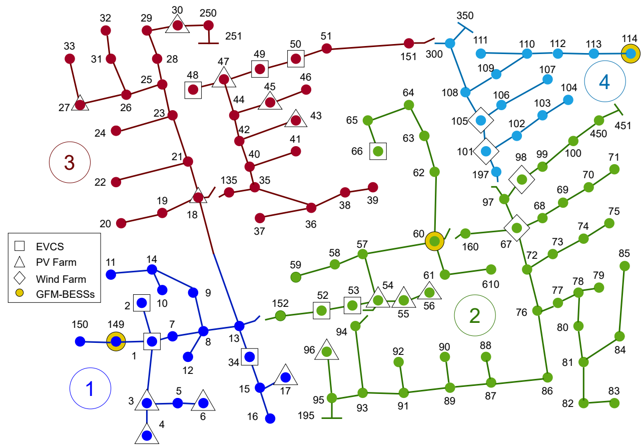

Fig. 3. Islanded IEEE 123 bus feeder with VPP zones [36].

TABLE I
SYSTEM AND DEVICE PARAMETERS.

Quantity	Symbol(s)	Value
Base power, frequency	S_{base}, f_0	15.7 MW, 50 Hz
GFM-BESSs (VSG / droop)	n_g, H_g	2 (2.0 s / 0.5 s)
Aggregate system inertia	H_{sys}	1.34 s
GFL agents, number of VPPs	$\lvert\mathcal{A}\rvert, \lvert\mathcal{Z}\rvert$	41 agents, 3 VPPs
Droop ceiling (BESS/V2G/DPV)	$K_{\max}$	0.30 / 0.20 / 0.15 $\cdot P_{\text{rated}}$
SoC band (Physical / FFR)	–	$[0.20, 0.90]$ / $[0.25, 0.85]$
Efficiency, V2G response lag	η, T_{v2g}	0.95, 0.3 s
Time step, Episode length	Δt	1 s / 300 steps

TABLE II
FREQUENCY-SECURITY THRESHOLDS [37], [34].

Quantity	Value
UFLS Stage 1 trip	49.0 Hz
Operator alarm / FFR target	49.5 Hz
UFLS Stage 1 delay	300 ms
RoCoF ride-through	2.0 Hz/s
FCR full activation	±200 mHz
Standard freq. range (settling band)	±50 mHz

TABLE III
CONTINGENCY SCENARIOS.

Sc.	Event	ΔP (MW)	% S_{base}	Purpose
S1	Load step	-2.5	16	Moderate
S2	Generation trip	-3.9	25	Severe under freq.
S3	Line trip	-2.4	15	Topology mutation
S4	High renewable surge	$+4.7$	30	Over freq. stress

B. Baselines and Training Protocol

The proposed controller is compared against five baselines (see Table IV). All learning baselines share the identical environment, observation, action, reward, and safety envelope of Sections II and III, so the comparison isolates the architecture rather than the safety set. MLP-MAPPO is the clean encoder ablation, replacing the GraphSAGE encoder by an MLP while holding the rest fixed; graph convolutional neural network (GCNN)-PPO and multi-agent twin delayed deep deterministic policy gradient (MATD3) are recent learning references that differ from the proposed method on more than one axis.

TABLE IV
BASELINES AND THE COMPONENT ISOLATED BY EACH METHOD.

#	Method	Encoder	RL algorithm	Learning architecture
1	GraphSAGE-MAPPO (*ours*)	GraphSAGE	PPO	MAPPO (CTDE)
2	MLP-MAPPO (ablation)	MLP	PPO	MAPPO (CTDE)
3	GCNN-PPO [38]	Spectral GCN	PPO	Centralized single-agent
4	MATD3 [39]	MLP	TD3	Multi-agent CTDE
5	Fixed droop	–	–	–
6	No FFR	–	–	–

Note: "–" indicates not applicable; RL: reinforcement learning; GCN: graph convolutional network; TD3: twin delayed deep deterministic policy gradient.

TABLE V
HYPERPARAMETERS SELECTED USING ASHA.

Hyperparameter	Symbol(s)	Value
Learning rate	α	$3.1 \times 10^{-5} \to 8 \times 10^{-6}$
Discount factor / GAE parameter	γ, λ	0.99, 0.95
PPO clipping parameter	ϵ	0.2
Entropy coefficient	c_{ent}	$5 \times 10^{-3} \to 0$
Embedding / hidden dimension	$d_{\text{emb}}, d_{\text{hid}}$	128/128
Number of GraphSAGE layers	L_{GNN}	2
Update epochs / minibatch size	N_{epoch}, B	4/16
Curriculum length / number of phases	$N_{\text{train}}, N_{\text{phase}}$	6000 episodes / 6 phases

Training uses a disturbance curriculum that escalates event magnitude and probability from load only events to the full $N-1$ mix at 40 % contingency magnitude. The selected configuration (see Table V) is shared by every learning baseline with identical capacity, optimizer, curriculum, and budget, so no method enjoys a tuning advantage and each is reported at convergence.

C. Evaluation Protocol

Each controller is evaluated zero-shot on the same 24 unseen reconfigurations, with the mean and a 95 % confidence interval reported over three seeds. Performance spans three metric groups: frequency security (nadir, windowed RoCoF, the integral of time-weighted absolute error (ITAE) as the primary settling metric, settling time, and UFLS avoidance rate), service and economics (committed reserve and revenue), and topology generalization (the training to unseen ITAE gap and its spread). The criteria follow the standards of Table II: RoCoF is checked against the 2.0 Hz/s ride-through. The proposed versus next best gap on the primary metrics is assessed by a paired Wilcoxon signed rank test.

V. RESULTS AND ANALYSIS

A. Frequency Security Performance

On the training topology G_0, Fig. 4 overlays the mean frequency trajectories of the six controllers. Because all RL-based methods operate under the same Lyapunov-certified safety envelope, they uniformly hold the post-event nadir and zenith at the alarm thresholds (see Table VI), avoiding any UFLS trip. The sawtooth on the proposed trace highlights the closed-form one-step projection activating at the band edge. Consequently, these comparable frequency bounds confirm that security is strictly enforced by the common safety layer, meaning any subsequent differences in ITAE, settling time, or topology transfer purely reflect the policy quality rather than unequal constraints.

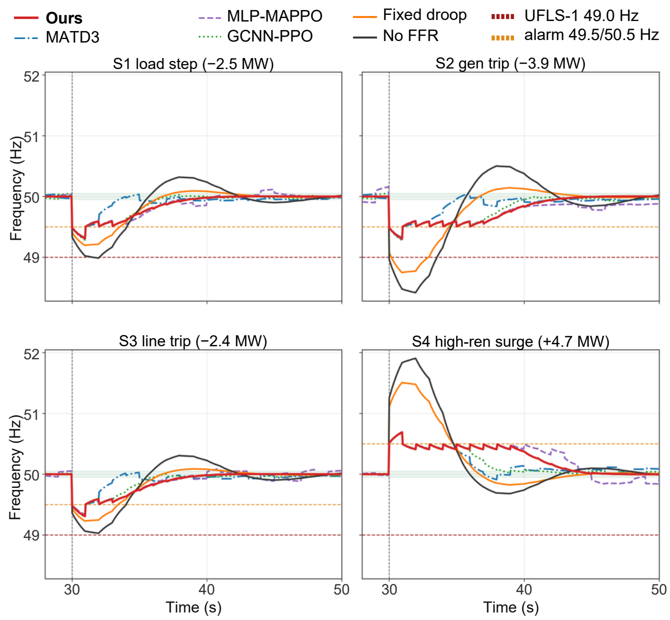

Fig. 4. Frequency response on the training feeder G_0.

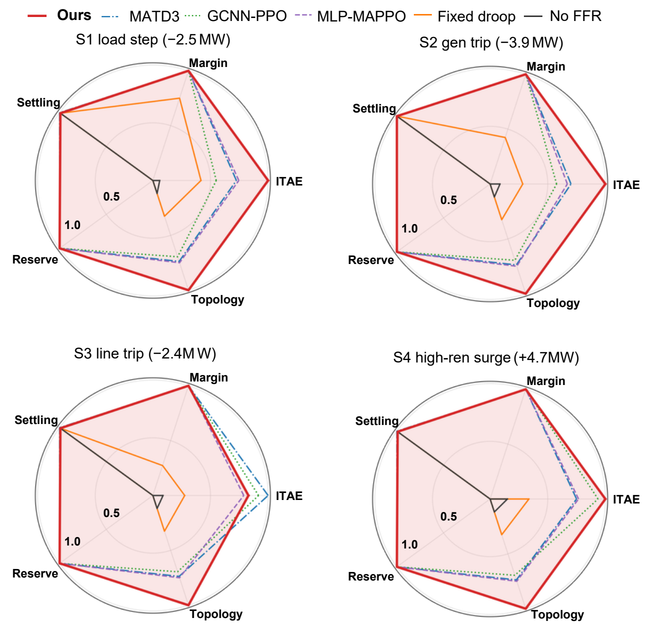

Fig. 5. Normalized per scenario performance radar across the 24 unseen topologies (larger is better).

TABLE VI
FREQUENCY SECURITY PERFORMANCE UNDER SEVERE EVENTS.

Method	S2 nadir (Hz)	S4 zenith (Hz)	Peak RoCoF (Hz/s)	RoCoF compliant	UFLS avoidance
GraphSAGE-MAPPO (*ours*)	49.50	50.50	1.24	✓	✓
MATD3 / GCNN-PPO / MLP-MAPPO	49.50	50.50	1.15–1.31	✓	✓
Fixed droop	48.75	51.51	2.23 / 2.69*	×	×
No FFR	48.42	51.91	2.61 / 3.15*	×	×

*Reported as S2/S4. Ride-through limit: 2.0 Hz/s.

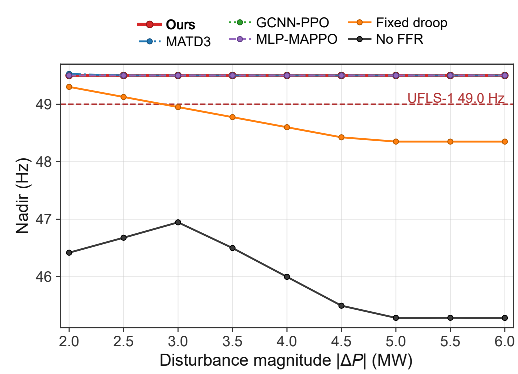

Fig. 6. Nadir versus disturbance magnitude over the 2–6 MW severity sweep.

Across the 24 unseen topologies, Fig. 5 scores the six controllers per contingency on five axes (post-event ITAE, nadir or zenith margin, settling speed, reliable reserve delivery, and topology robustness), oriented so that larger is better and normalized by the best in group value; the proposed controller encloses the largest area in every contingency.

Among the safe controllers, ITAE distinguishes two control philosophies. The saturating baselines commit reserve at the actuator limit regardless of the operating point, pulling the frequency back quickly but producing a slow, oscillatory recovery, whereas the proposed controller commits reserve selectively. On G_0, it achieves the lowest ITAE in three of the four contingencies, approximately 28% lower than that of the next-best learning controller for the load step and generation trip (e.g., 153 versus 213 $\text{Hz} \cdot \text{s}^2$ for S2), and remains within a few percent of the best result for the line trip. It returns to the ± 50 mHz band within 9–14 s, compared with 17–43 s for the baselines, and exhibits the smallest run-to-run variability.

UFLS avoidance and RoCoF compliance follow the same split (see Table VI): on the IEEE Std 1547 windowed RoCoF every learning controller respects the ride-through limit, while both non learning controls breach it on the severe events. This windowed quantity, unlike the instantaneous RoCoF set by the backbone, reflects the GFL damping over the step, which is what separates the learning controllers here.

The advantage scales with disturbance severity (see Fig. 6): across the 2–6 MW sweep, all learning controllers hold the nadir flat at the 49.5 Hz alarm, whereas fixed droop sinks below the 49.0 Hz UFLS-1 threshold beyond ~ 3 MW and the unregulated case falls steadily toward 45 Hz. This gap highlights the value of the joint power reference and droop gain action. While the fixed droop baseline relies on a static gain, causing insufficient support during severe events, the proposed policy jointly adapts both active power injection and frequency sensitivity online. The resulting disturbance-dependent reserve allocation keeps the nadir invariant across the entire range, proving that the dual-channel action space enables precise, adaptive state conditioning rather than a rigid single operating point calibration.

B. Topology Generalization

Topology generalization, the central claim of this work, is assessed using the post-event ITAE across 24 unseen topologies (see Fig. 7). GraphSAGE-MAPPO achieves the lowest mean unseen ITAE, 141.6 ± 36.6 $\text{Hz} \cdot \text{s}^2$, representing reductions of 31–43% relative to the learning baselines, 73% relative to fixed droop, and 91% relative to the unregulated case. It generalizes without performance loss: the unseen-topology mean is 7.4% lower than the value on G_0 (152.8 $\rightarrow$ 141.6), while the worst individual topology exhibits an increase of only 1.4%. The proposed controller outperforms all baselines on every holdout network (paired Wilcoxon signed-rank test, $p < 10^{-6}$).

This performance highlights the synergy between the joint action space and the inductive graph encoder. Unlike topology-blind baselines restricted to single-channel actions or fixed adjacencies (MLP/GCNN), the proposed joint power reference and droop gain coordination allows DERs to scale responses dynamically based on local connectivity, disturbance severity, and headroom. By reconstructing node embeddings directly from the reconfigured grid without retraining, the inductive architecture preserves real-time reserve-routing information, suppressing post-nadir oscillations to minimize ITAE.

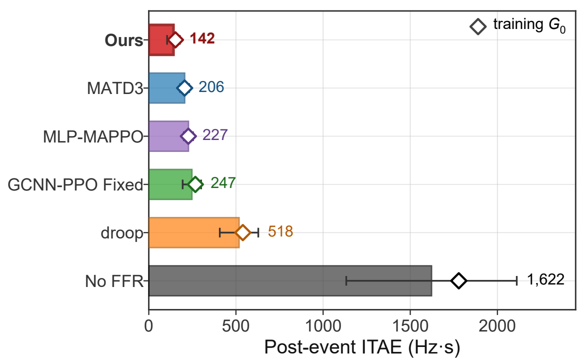

Fig. 7. Post event ITAE on the 24 unseen feeder topologies (lower is better; bar: unseen mean, whisker: spread across topologies, ◇: training G_0).

C. Ancillary Market Competitiveness

In least-cost FFR procurement, the per-event DSO cost corresponds to the price at which a controller's VPP offers the service; thus, a lower cost indicates a more competitive bid [40].

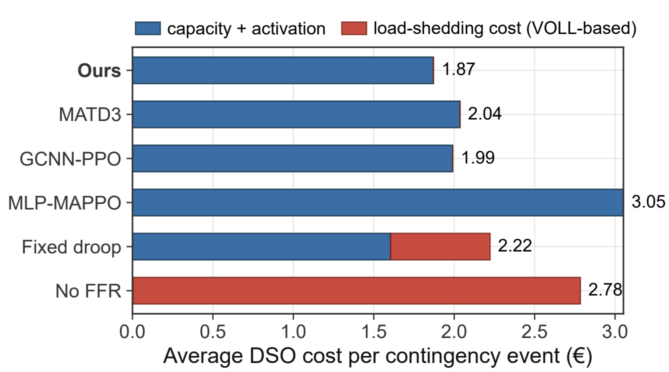

Fig. 8. Per-event DSO cost breakdown, averaged over the four contingencies within the 50-s settlement window: reserve service payments (capacity and activation) and VOLL-based load-shedding costs.

Across 24 unseen topologies, the proposed controller induces the lowest per-event DSO cost on every holdout network, outperforming the next-best compliant baseline in $24/24$ cases (paired Wilcoxon signed rank, $p = 1.2 \times 10^{-7}$). Its mean per-event cost is 10 % below GCNN-PPO, 13 % below MATD3, and 55 % below MLP-MAPPO. At the severe trip S2, all four learning controllers drive energy not served to zero; hence, the gap represents pure reserve service invariant to VOLL $\in \{1500, 3000, 8000\}$ €/MWh, whereas the non-learning controls incur substantial VOLL-based load-shedding costs (see Fig. 8). Normalized by handled imbalance energy, this optimal economic ordering holds across all contingencies (see Table VII).

This economic advantage supports the topology-coupled deliverability model. By conditioning reserve deployment on the active feeder graph, bus voltages, and inverter loading, the proposed framework avoids over-procuring network-inaccessible flexibility and directs physically deliverable reserve toward the disturbance location. The resulting cost reduction therefore reflects improved physical deliverability rather than a market artifact.

TABLE VII
MEAN NORMALIZED FFR PROCUREMENT COST (C/MWH) OVER $N = 10$ EPISODES PER SCENARIO

Method	S1	S2	S3	S4
Ours	$\mathbf{56.5 \pm 0.1}$	$\mathbf{37.0 \pm 0.0}$	$\mathbf{57.2 \pm 0.1}$	$\mathbf{28.8 \pm 0.1}$
GCNN-PPO	64.6 ± 0.3	41.5 ± 0.3	61.8 ± 1.1	34.6 ± 0.5
MATD3	64.2 ± 0.8	42.1 ± 0.7	69.0 ± 2.1	34.4 ± 0.7
MLP-MAPPO	127.3 ± 6.2	78.1 ± 2.1	128.7 ± 5.0	81.8 ± 2.7

Taken together, the results identify three complementary mechanisms: the safety envelope ensures frequency-security compliance, the joint power-reference and droop-gain action allocates reserve according to disturbance severity and resource availability, and the inductive GraphSAGE encoder preserves topology awareness after feeder reconfiguration. Their combination enables superior frequency performance, zero-shot topology robustness, and the lowest ancillary-service procurement cost.

VI. Conclusion

This paper addressed FFR in 100% inverter-based islanded microgrids under feeder reconfiguration, framing the controller as the topology-aware execution layer of a VPP providing FFR as an ancillary service. The topology coupled deliverability and frequency model is validated by the gap it exposes: under located forcing the realized COI RoCoF is 2.3–3.1 times lower than the uniform injection scalar bound, confirming that the active graph, not the aggregate inertia alone, governs how strongly a disturbance is felt. The proposed controller, with its joint power and droop action inside a shared safety envelope, holds every learning controller to a unit FFR success rate and full RoCoF compliance, while committing reserve selectively rather than saturating, which yields the lowest integral error of any standards-compliant controller. In zero-shot evaluation across 24 unseen reconfigurations, the proposed controller achieves the lowest mean post-event ITAE among all standards-compliant controllers, reducing ITAE by 1.14–1.75 times relative to the learning baselines and by 3.7 times relative to fixed droop. Its performance does not degrade beyond the training topology (-7.4% relative to G_0, with only a 1.4% increase on the worst-performing topology), while an encoder-only ablation attributes this gain to the inductive representation rather than to the cooperative RL stack. Economically, a VPP operating the proposed controller provides FFR at the lowest per-event procurement cost among all standards-compliant providers under a common administrative tariff, thereby yielding the most competitive offer under least-cost procurement. Future work will validate the proposed control framework on a commercial electromagnetic-transient platform to analysis nonlinear converter dynamics, fast oscillatory interactions.Futhermore hardware-in-the-loop testbed to analysis the effects of communication delays.

References

[1] N. K. Roy, S. Islam, A. K. Podder, T. K. Roy, and S. M. Muyeen, "Virtual inertia support in power systems for high penetration of renewables–Overview of categorization, comparison, and evaluation of control techniques," *IEEE Access*, vol. 10, pp. 129 190–129 216, 2022.

[2] Q. Salem, B. B. Fawaz, R. Aljarrah, and M. Karimi, "Grid forming converters for low inertia systems-capabilities and limitations: A critical review," *IEEE Open J. Ind. Electron. Soc.*, vol. 6, pp. 775–801, 2025.

[3] Q. Liu, R. Lyu, Z. Zhai, Y. Shen, X. Liu, and H. Guo, "Integrating fast-response capability into virtual power plant operation for ancillary services," in *2025 IEEE Kiel PowerTech*, 2025, pp. 1–6.

[4] O. Boutfarjoute *et al.*, "Fuzzy logic-based adaptive virtual inertia control for enhancing frequency stability in low-inertia microgrids," *Int. J. Power Electron. Drive Syst.*, vol. 16, no. 1, pp. 12–24, 2025.

[5] R. Jacob *et al.*, "Real-time outage management in active distribution networks using reinforcement learning over graphs," *Nat. Commun.*, vol. 15, p. 3456, 2024.

[6] W. M. Hamanah *et al.*, "Realization of robust frequency stability in low-inertia islanded microgrids with optimized virtual inertia control," *IEEE Access*, vol. 12, pp. 1–14, 2024.

[7] J. Köhler, R. Soloperto, M. A. Müller, and F. Allgöwer, "A computationally efficient robust model predictive control framework for uncertain nonlinear systems," *IEEE Trans. Autom. Control*, vol. 66, no. 2, pp. 794–801, 2021.

[8] W. Shi, D. Zhang, X. Han, X. Wang, T. Pu, and W. Chen, "Coordinated operation of active distribution network, networked microgrids, and electric vehicles: A multi-agent PPO optimization method," *CSEE J. Power Energy Syst.*, vol. 11, no. 5, pp. 1972–1985, 2025.

[9] D. Qiu, Y. Wang, T. Zhang, M. Sun, and G. Strbac, "Hierarchical multi-agent reinforcement learning for repair crews dispatch control towards multi-energy microgrid resilience," *Appl. Energy*, vol. 336, 2023.

[10] D. Wang, Q. Sun, and H. Su, "Real-time optimal energy management of microgrid based on multi-agent proximal policy optimization," *Neural Comput. Appl.*, vol. 37, no. 10, p. 7145–7157, Feb. 2025.

[11] Y. Wang, D. Qiu, Y. Wang, M. Sun, and G. Strbac, "Graph learning-based voltage regulation in distribution networks with multi-microgrids," *IEEE Trans. Power Syst.*, vol. 39, no. 1, pp. 1881–1895, 2024.

[12] C. Hu, Z. Cai, Y. Zhang, R. Yan, Y. Cai, and B. Cen, "A soft actor-critic deep reinforcement learning method for multi-timescale coordinated operation of microgrids," *Prot. Control Mod. Power Syst.*, vol. 7, 2022.

[13] M. Zhang, Y. Xu, L. Sang, Y. Guo, and N. D. Hatziargyriou, "Virtual power plants for frequency regulation: A learning-based method with safety guarantee," *IEEE Trans. Smart Grid*, vol. 17, pp. 388–401, 2026.

[14] J. Fernandopulle, L. Meegahapola, and I. Nutkani, "A VPP based techno-economic optimization framework for frequency stability enhancement," in *2025 IEEE PES 35th Australasian Universities Power Engineering Conference (AUPEC)*, 2025, pp. 1–5.

[15] Y. Pei *et al.*, "An emergency control strategy for undervoltage load shedding of power system: A graph deep reinforcement learning method," *IET Gener. Transm. Distrib.*, vol. 17, no. 14, pp. 3132–3145, 2023.

[16] C. Chen, Q. Li, L. Chen, Y. Liang, and H. Huang, "An improved GraphSAGE to detect power system anomaly based on time-neighbor feature," *Energy Rep.*, vol. 9, pp. 930–937, 2023.

[17] X. Hou, Y. Sun, X. Zhang, J. Lu, P. Wang, and J. M. Guerrero, "Improvement of frequency regulation in VSG-based AC microgrid via adaptive virtual inertia," *IEEE Trans. Power Electron.*, vol. 35, 2020.

[18] X. Xiong, X. Li, B. Luo, M. Huang, C. Zhao, and F. Blaabjerg, "An additional damping torque method for low-frequency stability enhancement of virtual synchronous generators," *IEEE Trans. Power Electron.*, vol. 39, no. 12, pp. 15 858–15 869, 2024.

[19] R. E. Alden, S. Silvestri, M. McCulloch, and D. M. Ionel, "EV charging and V2G operation for distribution system VPP including model predictive control," *IEEE Access*, vol. 14, 2026.

[20] C. B. Domenech, J. Naughton, S. Riaz, and P. Mancarella, "Towards distributed energy markets: Accurate and intuitive DLMP decomposition," *IEEE Trans. Energy Markets Policy Regul.*, vol. 2, 2024.

[21] H. Liu, J. Qiu, and J. Zhao, "A data-driven scheduling model of virtual power plant using Wasserstein distributionally robust optimization," *Int. J. Electr. Power Energy Syst.*, vol. 137, p. 107801, 2022.

[22] A. Arrigo, C. Ordoudis, J. Kazempour, Z. De Grève, J.-F. Toubeau, and F. Vallée, "Wasserstein distributionally robust chance-constrained optimization for energy and reserve dispatch: An exact and physically-bounded formulation," *Eur. J. Oper. Res.*, vol. 296, pp. 304–322, 2022.

[23] S. Huang, Q. Wu, S. S. Oren, R. Li, and Z. Liu, "Distribution locational marginal pricing through quadratic programming for congestion management in distribution networks," *IEEE Trans. Power Syst.*, vol. 30, no. 4, pp. 2170–2178, 2015.

[24] L. Luo and S. V. Dhople, "Spatiotemporal model reduction of inverter-based islanded microgrids," *IEEE Trans. Energy Convers.*, vol. 29, 2014.

[25] O. Ajala, N. Baeckeland, B. Johnson, S. Dhople, and A. Domínguez-García, "Model reduction and dynamic aggregation of grid-forming inverter networks," *IEEE Trans. Power Syst.*, vol. 38, 2023.

[26] Q. Shi, F. Li, and H. Cui, "Analytical method to aggregate multi-machine SFR model with applications in power system dynamic studies," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 6355–6367, 2018.

[27] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Int. Conf. Neural Information Processing Systems (NeurIPS)*, 2017, pp. 1024–1034.

[28] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. WU, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Curran Associates, Inc., 2022, pp. 24 611–24 624.

[29] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," 2018. [Online]. Available: https://arxiv.org/abs/1506.02438

[30] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017. [Online]. Available: https://arxiv.org/abs/1707.06347

[31] H. Alduaij and Y. Weng, "Reinforcement learning for stability-guaranteed adaptive optimal primary frequency control of power systems using partially monotonic neural networks," *IEEE Open Access J. Power Energy*, vol. 12, pp. 259–269, 2025.

[32] D. Liberzon, *Stability under Constrained Switching*. Boston, MA: Birkhäuser Boston, 2003, pp. 53–71.

[33] R. W. Kenyon, A. Sajadi, M. Bossart, A. Hoke, and B.-M. Hodge, "Interactive power to frequency dynamics between grid-forming inverters and synchronous generators in power electronics-dominated power systems," *IEEE Syst. J.*, vol. 17, no. 3, pp. 3456–3467, 2023.

[34] "IEEE standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces," *IEEE Std 1547-2018 (Rev. of IEEE Std 1547-2003)*, pp. 1–138, 2018.

[35] D. Tabas and B. Zhang, "Computationally efficient safe reinforcement learning for power systems," in *2022 American Control Conference (ACC)*, 2022, pp. 3303–3310.

[36] Y. Chai, L. Guo, C. Wang, Z. Zhao, X. Du, and J. Pan, "Network partition and voltage coordination control for distribution networks with high penetration of distributed PV units," *IEEE Trans. Power Syst.*, vol. 33, no. 3, pp. 3396–3407, 2018.

[37] ENTSO-E, "Frequency ranges: ENTSO-E guidance document for national implementation of frequency ranges for network codes on grid connection," ENTSO-E, Brussels, Belgium, Implementation Guidance Document, Apr. 2021.

[38] W. Guo, H. Du, T. Han, S. Li, C. Lu, and X. Huang, "Learning-driven load frequency control for islanded microgrid using graph networks-based deep reinforcement learning," *Front. Energy Res.*, vol. 12, 2024.

[39] J. Li and T. Zhou, "A robust large-scale multiagent deep reinforcement learning method for coordinated automatic generation control of integrated energy systems in a performance-based frequency regulation market," *IEEE/CAA J. Autom. Sinica*, vol. 12, pp. 1475–1488, 2025.

[40] R. Li, Q. Wu, and S. S. Oren, "Distribution locational marginal pricing for optimal electric vehicle charging management," *IEEE Trans. Power Syst.*, vol. 29, no. 1, pp. 203–211, 2014.